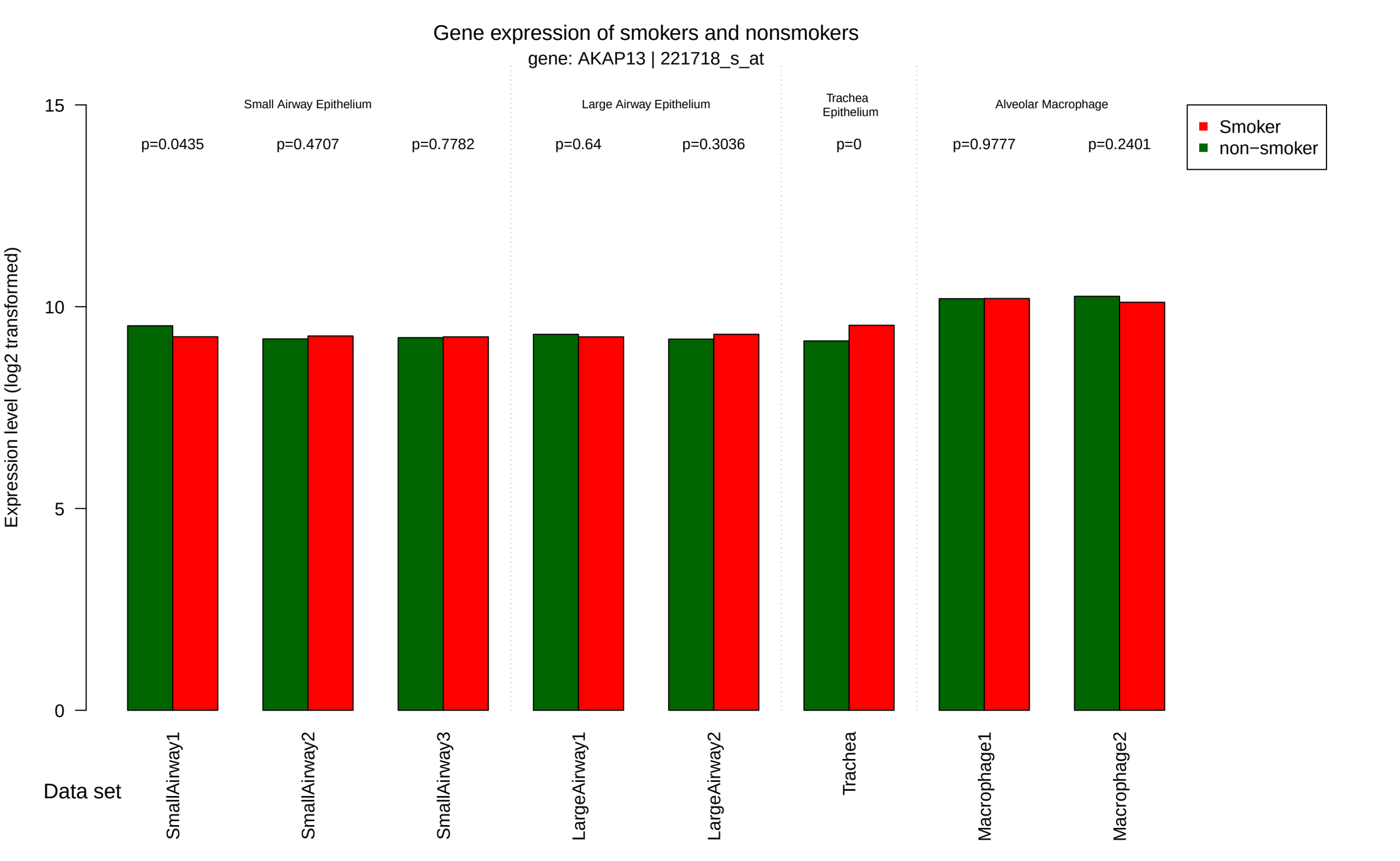
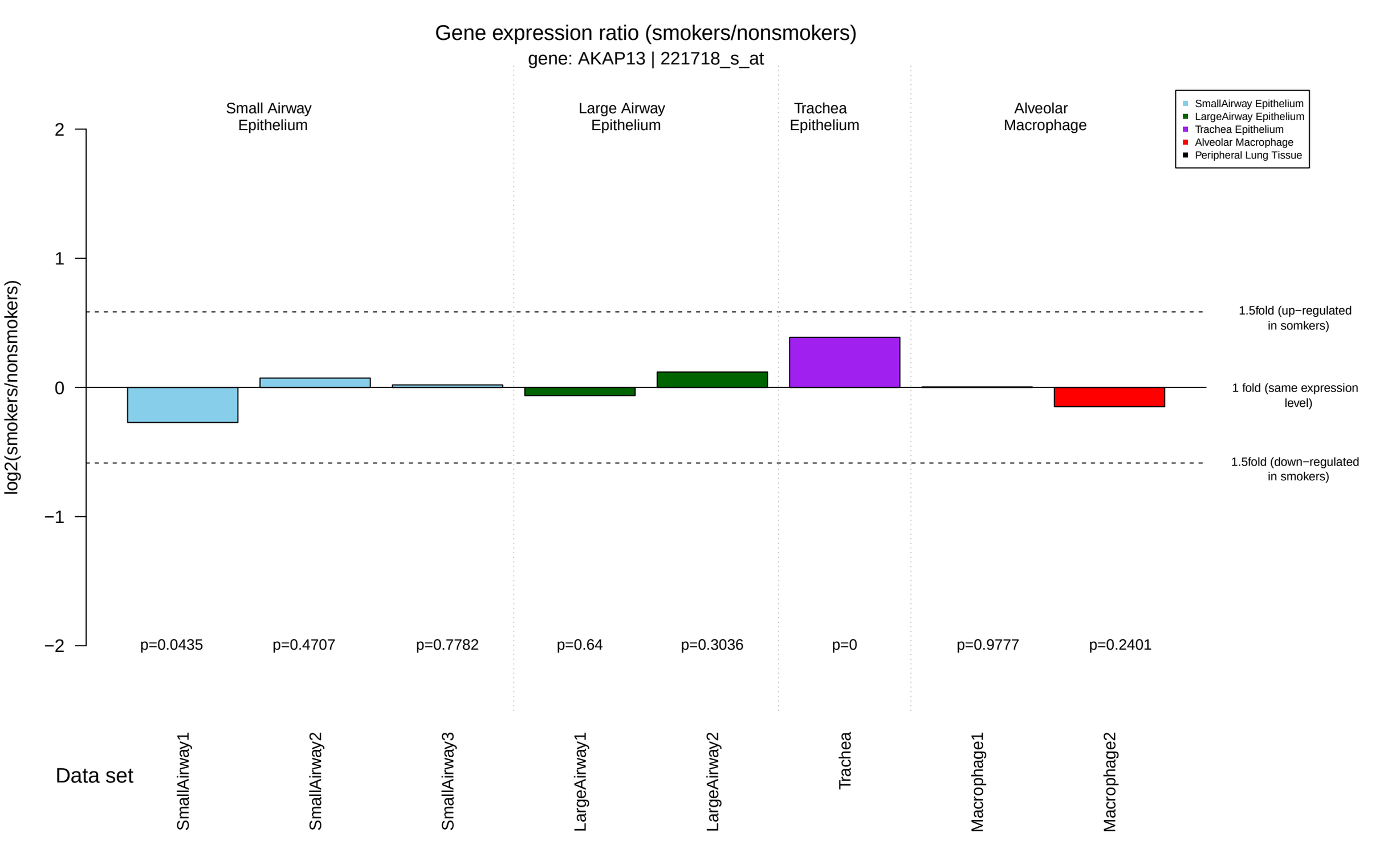
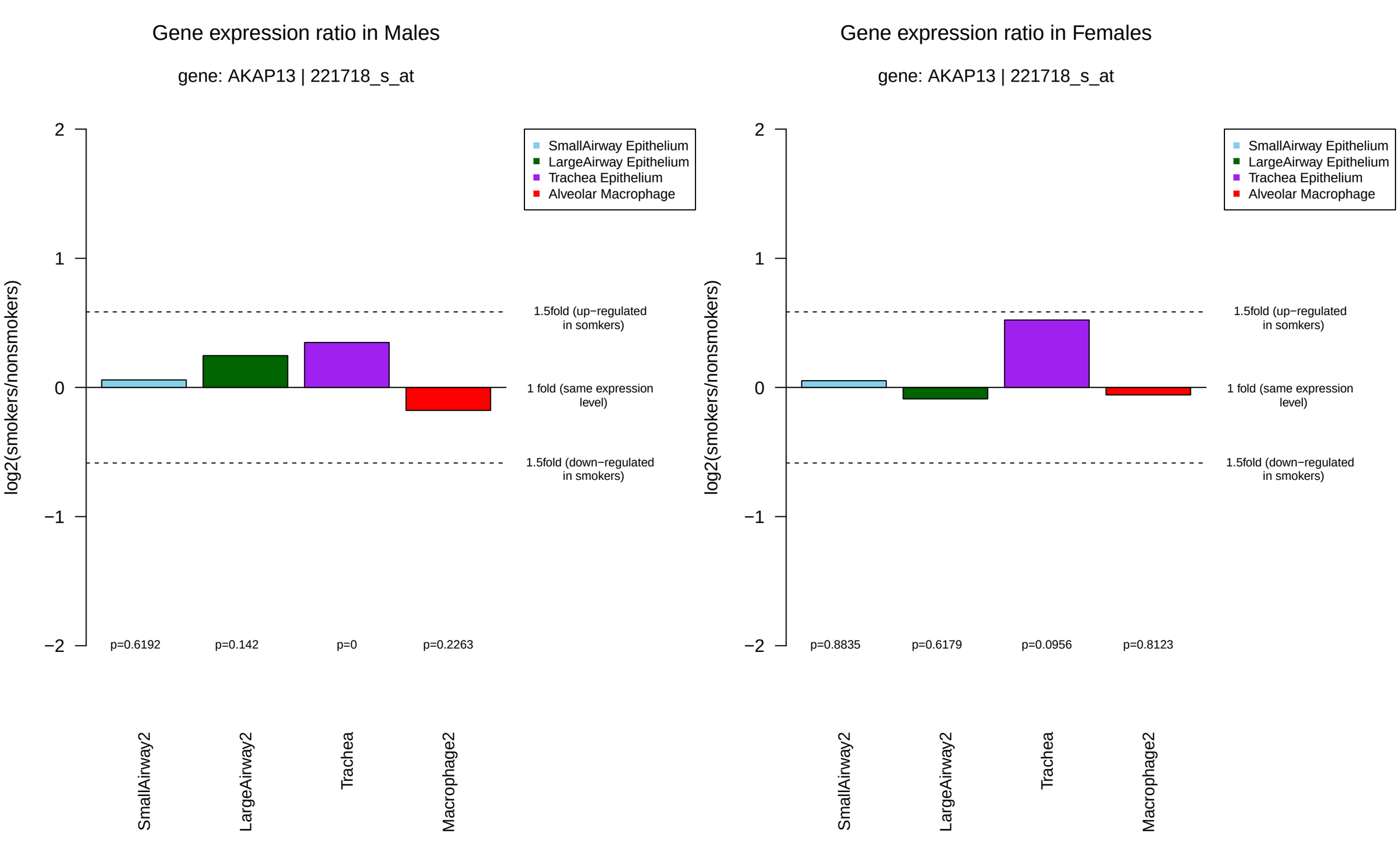
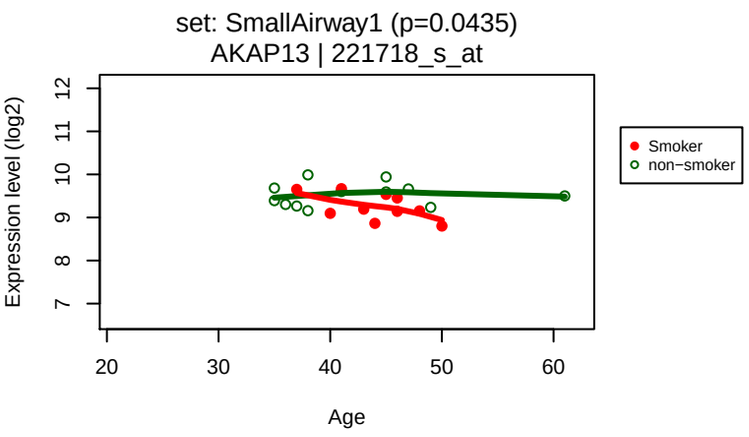
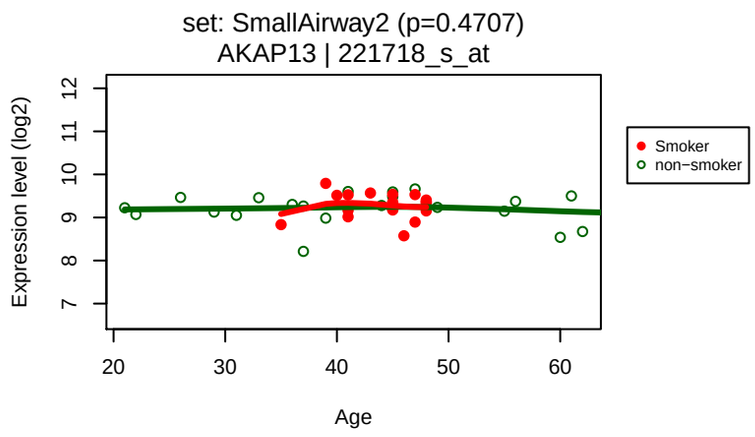
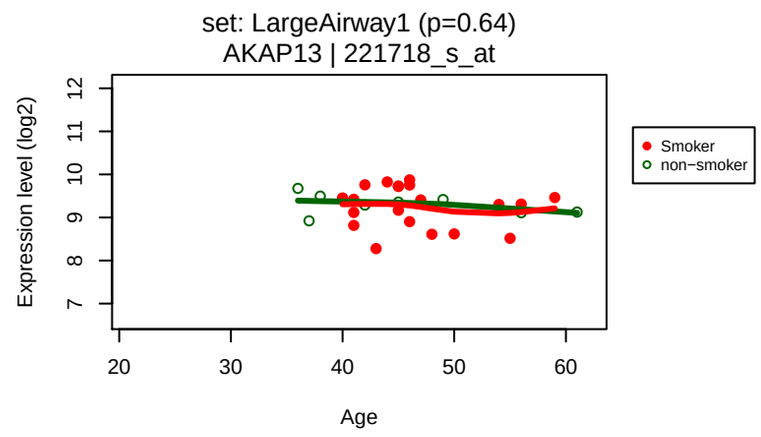

Gene expression of smokers and nonsmokers
gene: AKAP13 | 221718_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0435
p=0.4707
p=0.7782
p=0.64
p=0.3036
p=0
p=0.9777
p=0.2401
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

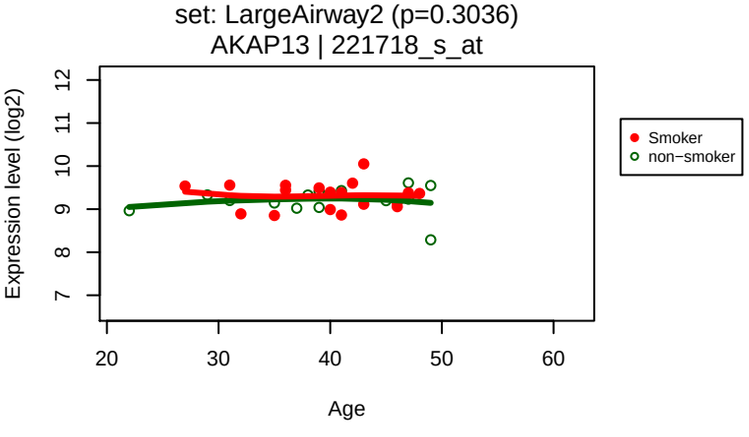

Gene expression ratio (smokers/nonsmokers)
gene: AKAP13 | 221718_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0435
p=0.4707
p=0.7782
p=0.64
p=0.3036
p=0
p=0.9777
p=0.2401
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

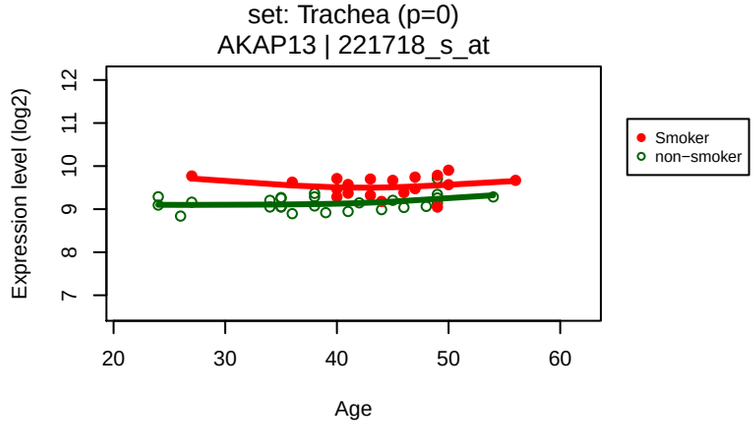

Gene expression ratio in Males
gene: AKAP13 | 221718_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6192
p=0.142
p=0
p=0.2263
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP13 | 221718_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8835
p=0.6179
p=0.0956
p=0.8123
SmallAirway2
LargeAirway2
Trachea
Macrophage2

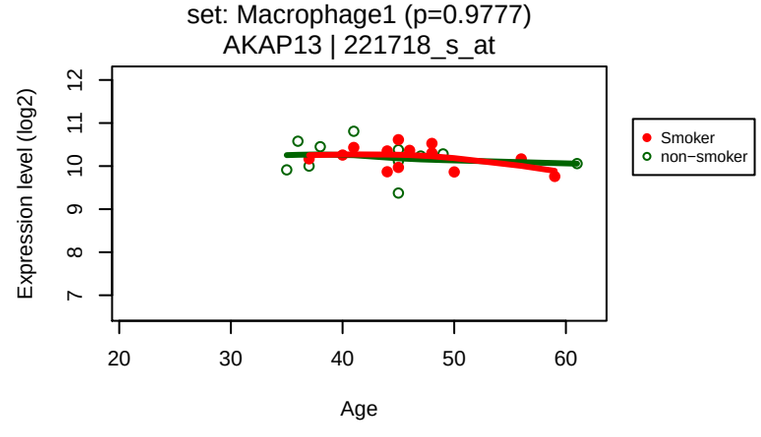

set: SmallAirway1 (p=0.0435)
AKAP13 | 221718_s_at
Expression level (log2)
Age
Smoker
non−smoker

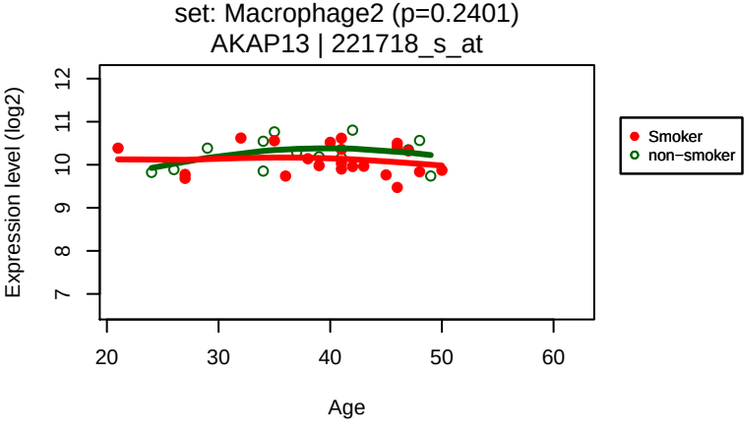

set: SmallAirway2 (p=0.4707)
AKAP13 | 221718_s_at
Expression level (log2)
Age
Smoker
non−smoker

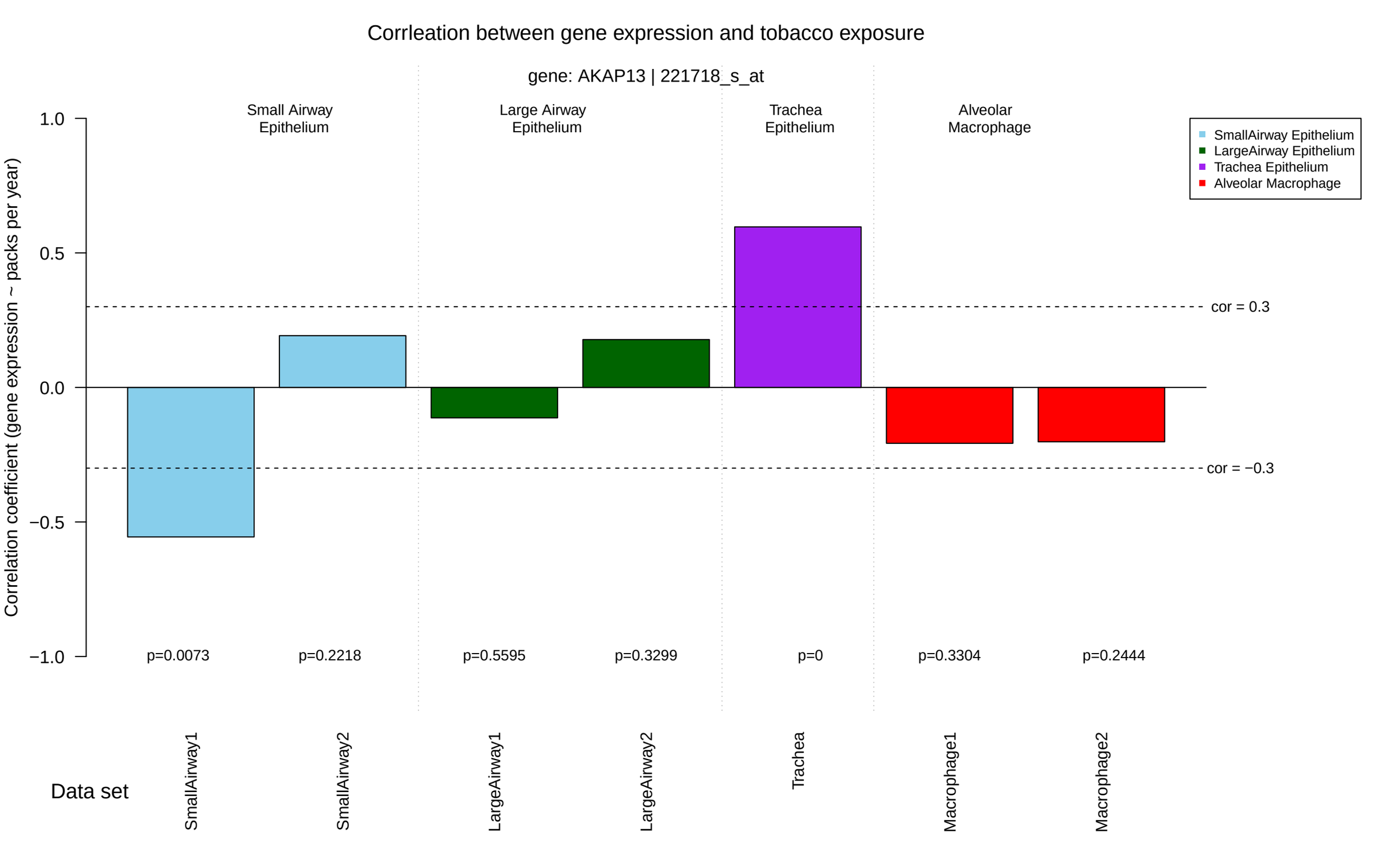

set: LargeAirway1 (p=0.64)
AKAP13 | 221718_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3036)
AKAP13 | 221718_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
AKAP13 | 221718_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9777)
AKAP13 | 221718_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2401)
AKAP13 | 221718_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP13 | 221718_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0073
p=0.2218
p=0.5595
p=0.3299
p=0
p=0.3304
p=0.2444
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2